NNN-G-692C
March 6, 1970
SUPERSEDING
Int. Fed. Spec. NNN-G-00692B(GSA-FSS)
June 4, 1963 and
Fed. Spec. NNN-G-692A
March 29, 1963

FEDERAL SPECIFICATION

GRADUATE, LIQUID, LABORATORY AND PHARMACEUTICAL, GLASS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass graduates of a grade suitable for laboratory, pharmaceutical, and other uses.

1.2 Classification.

1.2.1 Types. Graduates shall be of the following types, as specified (see 6.1).

Type I - Graduated in metric scale.
Type II - Graduated in apothecary scale.
Type III - Graduated in both metric and apothecary scales.

1.2.2 Capacities. Graduates shall be of the following capacities and graduated as indicated for the type specified (see 6.1).

Type I	Type II	Type III
5 milliliters (ml.)	60 minims	--------
10 ml.	120 minims	--------
--------	4 fluid drams (fl.drs.)	--------
25 ml.	8 fl. drs.	25 ml. and 8 fl. drs.
50 ml.	2 fluid ounces (fl.ozs.)	50 ml. and 2 fl. ozs.
100 ml.	4 fl. ozs.	100 ml. and 4 fl. ozs.
250 ml.	8 fl. ozs.	250 ml. and 8 fl. ozs.
500 ml.	16 fl. ozs.	500 ml. and 16 fl. ozs.
1000 ml.	32 fl. ozs.	1000 ml. and 32 fl. ozs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

DD-G-541 - Glass (Laboratory).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

FSC 6640

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids shall apply.

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Tariff Order Section. 1616 P Street. N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

American Society for Testing and Materials (ASTM) Standards:

C 148-65 Polariscopic Examination of Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The glass shall conform to the requirements of type I, class A of DD-G-541.

3.1.1 Annealing. Ring strain shall be no greater than Temper No. 1 when tested as specified in 4.4.3. Any longitudinal strain shall be faint and highly diffuse.

3.2 Design.

3.2.1 Style. Graduates of a capacity of more than 4 fluid drams (15 ml.) shall be conical. Graduates of a capacity of 4 fluid drams or less shall be cylindrical.

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3.2.2 Proportions. The inside measurement from the bottom of the graduate to the capacity graduation shall be not less than 5 times the inside diameter on a cylindrical graduate and 2 times the inside diameter at the capacity graduation on a conical graduate. The inside measurement from the bottom of the conical graduate to the point representing one fourth of the capacity shall be not less than the inside diameter at that point.

3.2.3 Pouring lip. All graduates shall have a suitable pouring lip (see 3.2.5.2), and shall pour evenly when tested as specified in 4.4.1.

3.2.4 Base. The base shall be perpendicular to the vertical axis of the graduate and of such size that the empty graduate will stand on a surface making an angle of 25 percent or approximately 15 degrees with the horizontal.

3.2.5 Graduation markings. Each graduate shall have clear graduation markings, perpendicular to the vertical axis of the graduate and parallel to each other. Graduations shall be etched and shall not be wider than 0.015 inch (0.38 millimeter). No graduation shall extend less than one fourth the distance around the graduate. Graduates shall have an initial interval that is not subdivided, equal to not less than one fifth and not more than one fourth of the capacity of the graduate. Except for this initial interval, the values of all graduated intervals shall be the same. Each main graduation shall be marked to show its value; intermediate graduations shall not be marked. Value figures shall be uniformly positioned, either directly upon or immediately above the graduations to which they refer. Figures placed upon graduations shall be set in from the ends of the graduation a sufficient distance to allow the ends of the graduations to be used in making a setting.

3.2.5.1 Single scale graduates. On single scale graduates, type I and type II, the main graduations shall completely encircle the graduate. Intermediate graduations shall extend at least one half the distance around the graduate.

3.2.5.2 Double scale graduates. On double scale graduates, type III, there shall be a clear space between the ends of the main graduations of the two scales. This space shall be approximately 90 degrees from the pouring lip of the graduate and shall conform to the requirements of table I.

TABLE I. Clear space between ends of main graduations on double scale graduates.

Inside diameter of graduate at the graduation	Clear space between ends of main graduations
Inches	Inch
Less than 1-1/2	1/8 to 1/4
1-1/2 to 3, inclusive	1/4 to 1/2
Over 3	3/8 to 5/8

3.2.5.3 Marking details. Capacities, graduation ranges, values of graduated intervals, and numbered graduations, applicable to single scale graduates and to appropriate portions of double scale graduates shall be as shown in table II. Graduation markings shall be for 20°C.

TABLE II. Design details for graduates

Capacity	To be graduated between	Value of graduated intervals	To be numbered at each even
60 minims	15 and 60 minims	5 minims	[a]10 minims
120 minims	30 and 120 minims	10 minims	[b]20 minims
4 fl. drs.	1 and 4 fl. drs.	1/2 fl. dr.	1 fl. dr.
8 fl. drs.	2 and 8 fl. drs.	1 fl. dr.	2 fl. drs.
2 fl. ozs.	1/2 and 2 fl. ozs.	1/4 fl. oz.	1/2 fl. oz.
4 fl. ozs.	1 and 4 fl. ozs.	1/2 fl. oz.	1 fl. oz.
8 fl. ozs.	2 and 8 fl. ozs.	1/2 fl. oz.	1 fl. oz.
16 fl. ozs.	4 and 16 fl. ozs.	1 fl. oz.	2 fl. ozs.
32 fl. ozs.	8 and 32 fl. ozs.	2 fl. ozs.	4 fl. ozs.
5 ml.	1 and 5 ml.	1/2 ml.	1 ml.
10 ml.	2 and 10 ml.	1 ml.	2 ml.
25 ml.	5 and 25 ml.	5 ml.	5 ml.
50 ml.	10 and 50 ml.	5 ml.	10 ml.
100 ml.	20 and 100 ml.	10 ml.	20 ml.
250 ml.	50 and 250 ml.	25 ml.	50 ml.
500 ml.	100 and 500 ml.	25 ml.	50 ml.
1000 ml.	200 and 1000 ml.	50 ml.	100 ml.

a-and, in addition, at the first or 15 minims graduation.
b-and, in addition, at the first or 30 minims graduation.

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3.2.5.4 Permanency of markings. Graduation markings shall be permanent and legible. Graduation markings or identification markings shall be applied by one of the following methods -- etched and filled with a permanent pigment; by application of an enamel fired onto the glass without etching; or by application of the stain fired into the glass without etching. The appearance shall not change when tested for permanency of marking as specified in 4.4.2.

3.2.5.5 Tolerances. Tolerances shall be as shown in table III for graduates that are graduated "to deliver". The tolerance, to be applied at any graduation, is determined by the inside diameter of the graduate at the graduation in question. Diameters between the ranges shown in table III, such as 30.5 mm., shall be rounded to the nearest mm. and the appropriate tolerance used.

3.3. Identification marking. Graduates shall be permanently and legibly marked with the name or trademark of the manufacturer, the letters "TD", and the temperature of calibration as 20° C.

3.4 Workmanship. Graduates shall be symmetrical in shape, of uniform thickness, and free from checks, chips, streaks, stones, and bubbles, which detract from their appearance or may impair their serviceability.

TABLE III. Tolerances for graduates at 20° C.

Inside diameter of graduate--inches	± Tolerance minims	Inside diameter of graduate--mm.	± Tolerance ml.
1/2 or less	2	15 or less	0.1
9/16 to 3/4 inclusive (incl.)	3	16 to 20 incl.	0.2
13/16 to 1 incl.	6	21 to 25 incl.	0.4
1-1/16 to 1-1/4 incl.	10	26 to 30 incl.	0.6
1-5/16 to 1-1/2 incl.	15	31 to 35 incl.	0.8
1-9/16 to 1-3/4 incl.	20	36 to 40 incl.	1.1
1-13/16 to 2 incl.	30	41 to 45 incl.	1.4
2-1/16 to 2-1/4 incl.	40	46 to 50 incl.	1.8
2-5/16 to 2-1/2 incl.	50	51 to 55 incl.	2.2
2-9/16 to 2-3/4 incl.	65	56 to 60 incl.	2.8
2-13/16 to 3 incl.	80	61 to 65 incl.	3.4
3-1/16 to 3-1/4 incl.	95	66 to 70 incl.	4.1
3-5/16 to 3-1/2 incl.	110	71 to 75 incl.	4.8
3-9/16 to 3-3/4 incl.	130	76 to 80 incl.	5.6
3-13/16 to 4 incl.	150	81 to 85 incl.	6.4
		86 to 90 incl.	7.2
		91 to 95 incl.	8.1
		96 to 100 incl.	9.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.1), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificate of analysis. Certified test data, in lieu of contractor performing tests, will be acceptable as evidence of compliance with requirements of 3.1.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

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	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the graduate.

Characteristic	Requirement	Test procedure	Inspection level
Annealing	3.1.1	4.4.3	S-1
Pour test	3.2.3	4.4.1	S-1
Permanency of marking	3.2.5.4	4.4.2	S-1
Capacity	3.2.5.5		S-1

4.3 Examination. The graduates shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects *

Categories	Defects
Critical	Not defined.
Major	
101	Graduate not free of cracks.
102	Graduate not free of chips.
103	Graduation and calibration markings not complete, correct, or legible.
104	Graduate not free of sharp edges.
105	Graduate not free of protrusions.
106	Graduate not free of deformation.
Minor	
201	Graduate not free of stones and striae.
202	Graduate not free of bubbles.
203	Manufacturer's identification not complete, correct, or legible.

*Examination shall not be restricted to the defects listed above.

4.3.2 Dimensional examination. The graduates shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Level A and level B preparation for delivery (section 5).

Categories	Defects
Critical	Not defined.
Major	AQL 1.5 percent defective.
101	Unit, intermediate, or shipping containers not as specified.
102	Unlike graduates intermediately packaged or packed.
103	Unit, intermediate or shipping containers damaged.
104	Marking of unit, intermediate, or shipping container illegible, incorrect or incomplete.
105	Blocking or cushioning not as specified.

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4.4 Tests. Tests shall be conducted as follows.

4.4.1 Pour test. Fill the graduate to indicated capacity with water. Pour approximately one-fourth of the water from the graduate at such a rate that the water flows in a continuous stream and falls free from the lip of the graduate. If the stream of water clings to the outside of the graduate and flows off the bottom, it shall be considered cause for rejection.

4.4.2 Tests for permanency of color in etched or engraved markings. A different sample shall be used for each test specified below.

4.4.2.1 Resistance to alkali. The graduate shall be immersed completely in 1N sodium hydroxide at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The graduate should then be removed from the alkali, thoroughly rinsed in distilled water, and dried by rubbing vigorously with a cloth.

4.4.2.2 Resistance to acid.

4.4.2.2.1 Hydrochloric acid. The graduate shall be immersed completely in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, boiled for 1/2 hour, and allowed to stand for 1 hour in the acid. The graduate shall then be removed, thoroughly rinsed with distilled water and dried by rubbing vigorously with a cloth.

4.4.2.2.2 Sulfuric acid. The graduate shall be immersed completely in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The graduate shall then be removed, thoroughly rinsed with distilled water and dried by rubbing vigorously with a cloth.

4.4.3 Annealing. Annealing shall be tested according to ASTM C-148, Polariscopic Examination of Glass Containers.

4.5 Acceptance/rejection criteria. If any graduate fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each graduate shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636, type CF, class domestic, or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to insure against shifting or damage to the graduate. Box closure shall be secured with tape conforming to PPP-T-45.

5.1.1.2 Intermediate packaging. The number of graduates specified (see 6.1) shall be intermediately packaged in a box conforming to PPP-B-636, type CF, class domestic. Box closure shall be in accordance with the appendix of the box specification. All graduates in an intermediate package shall be of the same type and capacity.

5.1.2 Level C. Graduates shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The number of graduates specified (see 6.1), shall be packed in a box conforming to PPP-B-621, class 2; or PPP-B-601, overseas type. The container shall be provided with a liner conforming to MIL-L-10547, which shall be closed and sealed in accordance with the appendix thereto. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 Level B. The number of graduates specified (see 6.1), shall be packed in accordance with 5.2.1, except that shipping containers shall conform to PPP-B-621, class 1, PPP-B-601, domestic type; or PPP-B-636, type CF or SF, class domestic, grade not less than 175. Case liners shall not be required.

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5.2.3. Level C. Graduates shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and capacity required.
(c) Level of packaging and packing required.
(d) Number of graduates per intermediate and shipping container required.
(e) Special markings as required (see 5.3.1 and 5.3.2).

6.2 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

MILITARY INTEREST:

Custodians.

Army - MU
Navy - MS

Review activities:

Army - MD
Navy - MS
DSA - DP

User activities:

Army - MI, SM

Preparing activity:

GSA-FSS

CIVIL AGENCIES INTEREST:

AGR
COM-NBS
GSA-FSS
HEW
VA-DMS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. **Price 15 cents each.**

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment, detach, complete, and mail to General Services Administration, FSS (FMSC), Washington, D. C. 204(

NOTE Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

NNN-G-692C, Graduate, Liquid, Laboratory and Pharmaceutical, Glass

2 CONTRACT NO *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5 GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6 SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g tests in 4.2.2 will not assure that the battery will last required time, temperature ranges in table 2 do not conform to commercially available items)*

8. RECOMMENDATIONS

9 NAME OF MANUFACTURER, ASSOCIATION, GOVT. AGENCY, ETC	10 ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12 DATE

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